DESIGNING THE SCARE: INTERACTIVE FRIGHT EFFECTS WITH TOUCHDESIGNER & PYTHON

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ARDUINO IDE

TOUCHDESIGNER

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SECTION 1

Opening and Haunted Smelter Case Study

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THE HAUNTED SMELTER

Haunted Smelter Experience
Quincy Smelter – Hancock Michigan

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MOTION TRIGGER SCREEN

Room 1 – Multi-video motion triggered jump scare.

↓ PIR Sends Notification to Arduino
↓ Arduino receives and sends Serial message to TouchDesigner
→ TouchDesigner sends projection and sound out.

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PERFORMER TRIGGER SCREEN

Room 2 – Single Video with Lighting and Sound

↓ Trigger Notification to Arduino
↓ Arduino receives and sends Serial message to TouchDesigner
→ TouchDesigner sends projection and OSC commands to lighting and sound.

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BUILDING OUR OWN

Arduino Uno	Button, PIR motion sensor, and pressure pad as inputs
TouchDesigner	Reactive engine handling video, audio, and lighting output
Python in TD	Scripts inside nodes to interpret and route incoming data

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SECTION 2

Touchdesigner Orientation

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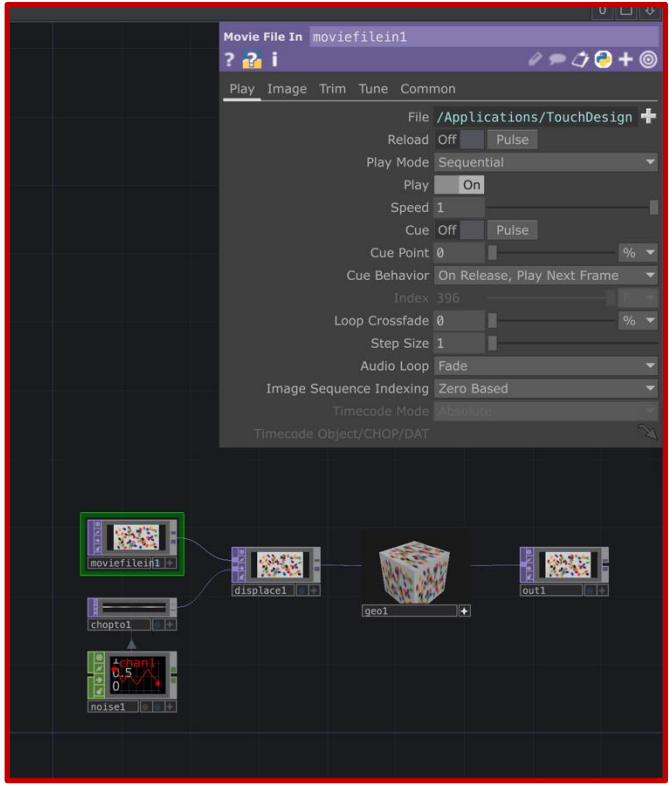
WHAT IS TOUCHDESIGNER?

- Node-based visual programming environment developed by Derivative
- Widely used in live entertainment, installation art, and interactive media
- Build by connecting operators — no traditional coding required to start
- Python available inside nodes for logic and control
- Used today to receive sensor data and drive scare outputs



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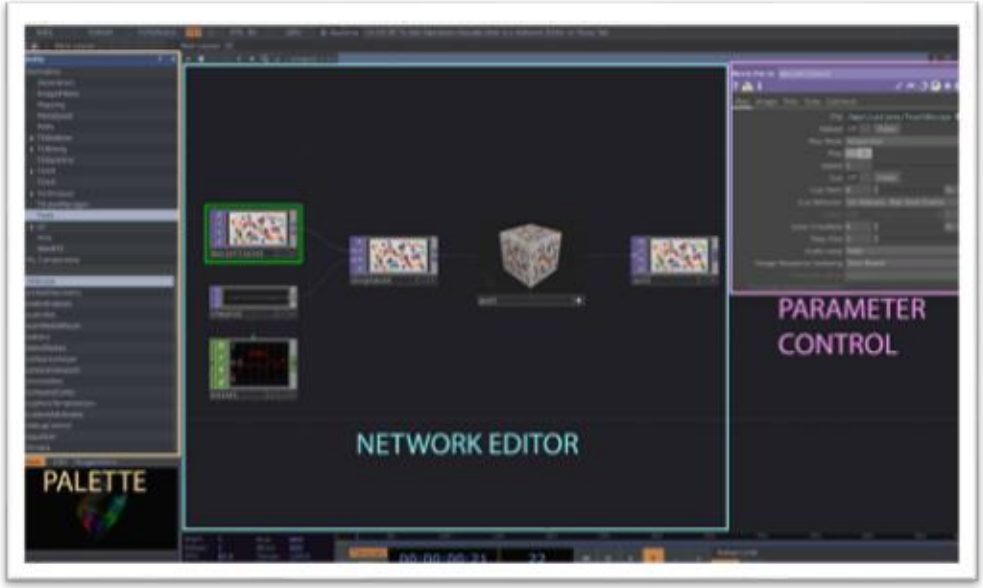
NAVIGATING THE INTERFACE



Network Editor: Build and connect operators here

Palette: Find and add operators

Parameters Panel: Control behavior of selected operator

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OPERATOR FAMILIES OVERVIEW

TOPs	CHOPs	DATs
Texture Operators	**Channel Operators**	**Data Operators**
Images and video	Time-based numeric data	Text, tables, scripts
★ Used today	★ Used today	★ Used today

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TOPS – Texture Operators

- Handle images and video — the visual output side of the system
- Movie File In TOP plays back video files
- TOPs pipe output to a display or projection surface
- Everything the audience sees passes through TOPs
- In our build: the scare video output

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CHOPS – Channel Operators

- Handle time-based numeric data — signals, values, triggers
- Sensor data from Arduino gets converted into CHOP channels
- CHOPs let you threshold, smooth, and map incoming data
- This is where trigger logic lives — sensor crosses a threshold, event fires
- In our build: routing Arduino sensor values to outputs

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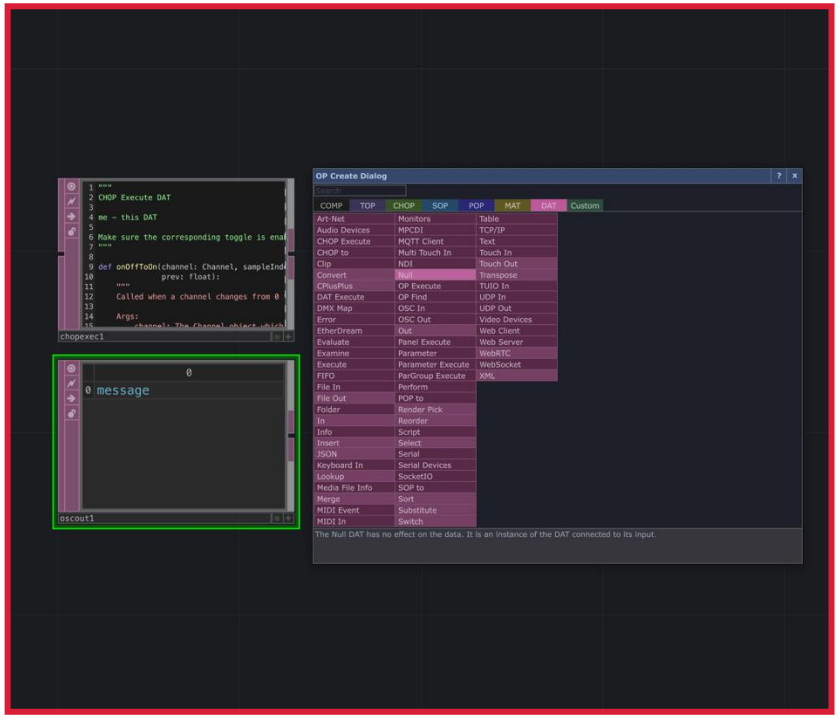
DATS – Data Operators

- Handle text, tables, and scripts
- Serial DAT reads the incoming data stream from Arduino
- Python scripts are written inside DAT nodes
- DATs are the nervous system — receiving and interpreting incoming information
- In our build: serial communication and Python trigger logic



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SECTION 3

Arduino Setup & Serial Communication

ARDUINO IDE

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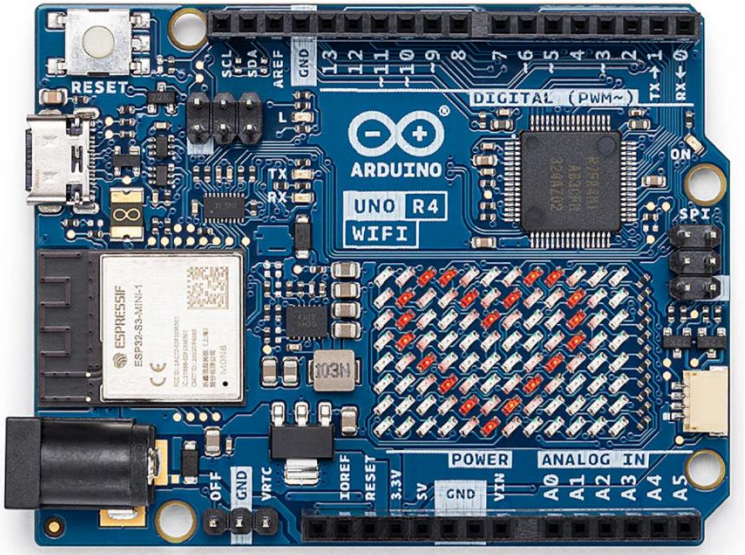
WHAT IS ARDUINO

- Open-source electronics platform combining a microcontroller board and programming environment
- Reads inputs from the physical world — sensors, buttons, switches — and converts them to data
- Sends that data via serial communication to other software — in our case, TouchDesigner
- The Arduino Uno is one of the most beginner-friendly boards availableKey areas on the board:
  - Power pins
  - Digital pins
  - Analog pins
  - USB connection



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CONFIRMING THE SERIAL CONNECTION

- Trigger each input — button, PIR, pressure pad — and confirm distinct values in the Serial Monitor
- Troubleshooting: verify the correct COM port is selected and baud rate matches the sketch
- If you see data in the Serial Monitor, your Arduino is working.

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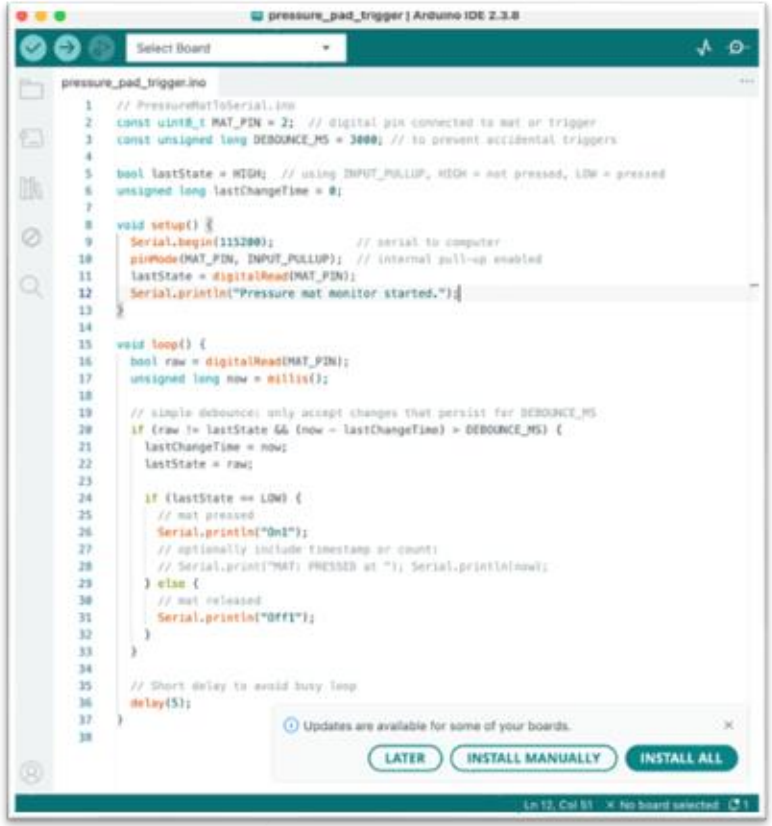
ARDUINO SKETCH – LINE BY LINE

- Variables – located at top
- setup() — declare pin modes
- Serial.begin() — open comms channel
- loop() — read sensor states
- Serial.println() — send value to TD
- Each input sends a distinct value



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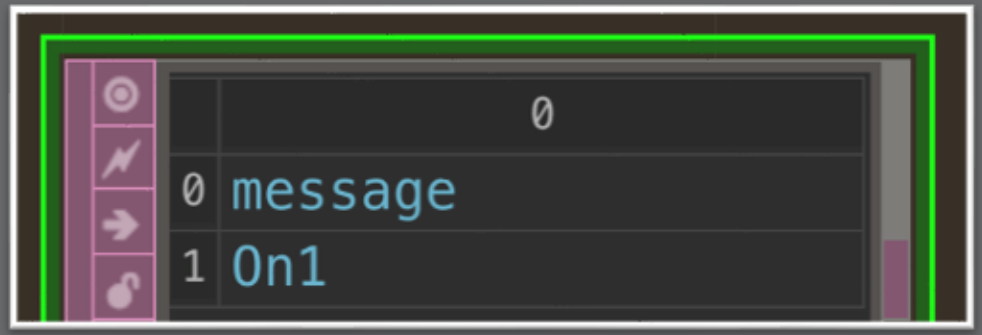
TRANSITION TO TOUCHDESIGNER

Close the Serial Monitor before switching to TouchDesigner.

> Only one application can read the serial port at a time — if the Serial Monitor is open, TD cannot connect.

We have data coming out of the Arduino. Now we need TouchDesigner to listen for it.



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SECTION 4

Follow-Along build: Sensors to Scare

TOUCHDESIGNER

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ORIENTATION TO STARTER FILE

- Input networks, one per sensor, that feed into an output layer
- Some nodes are pre-built; key connections and scripts are left for us to complete
- Goal: build one complete input-to-output chain, then replicate for the other two input

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BUILDING THE FIRST CHAIN: BUTTON INPUT

- Serial DAT confirms incoming Arduino data
- DAT Execute — Python script reads the value, identifies which input fired, and sets a trigger state
- Constant CHOP holds the trigger value
- Output layer: Movie File In TOP → video, Audio File In CHOP → sound, OSC Out DAT → lighting cue
- Test — press button, confirm all three outputs fire

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PYTHON INSIDE TOUCHDESIGNER: A BRIEF PAUSE

- Python inside TD nodes is the same language used everywhere — written inside a DAT
- No separate development environment requiredConcepts covered in this script:
  - Reading incoming serial data
  - Conditional logic — which sensor fired?
  - Setting trigger state values
- Arduino sends data → Python interprets it → TD responds

```python
def ignoreForSeconds(seconds):
    """
    Call this function to ignore messages for a specified duration
    """
    me.store('ignoreUntil', time.time() + seconds)
    print(f"Ignoring serial messages for {seconds} seconds")

def stopIgnoring():
    """
    Call this to immediately stop ignoring messages
    """
    me.store('ignoreUntil', 0)
    print("No longer ignoring serial messages")

def onTableChange(dat):
    # Fetch ignoreUntil time (default 0)
    ignoreUntil = me.fetch('ignoreUntil', 0)
    print(ignoreUntil)
    currentTime = time.time()
    print(currentTime)

    # If still in ignore period, exit early
    if currentTime < ignoreUntil:
        return

    # Process only if there's at least one data row
    if dat.numRows > 1:
        message = dat[1, 0].val.strip()
datexec1
```

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REPLICATING FOR PIR MOTION SENSOR

- Logic is identical — only the incoming value from the Arduino changes
- Update the Python script to respond to the PIR sensor value
- About PIR sensors:
  - Passive infrared — detects body heat and movement
  - No interaction required from the guest
  - Standard in haunted attractions for concealed detection
- Test — trigger PIR and confirm all outputs fire

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REPLICATING FOR PRESSURE PAD

- Same process — update the Python script for the pressure pad valueAbout pressure pads:
  - Hidden triggers — placed under rugs, floors, or props
  - Guest controls timing without realizing it
  - Useful for spacing guest groups in a walkthrough
- Test — trigger pressure pad and confirm all outputs fire

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TESTING THE FULL SYSTEM

- Trigger each input in sequence — confirm the system responds correctly to all three
- Cooldown logic: in a real attraction, a timer prevents retriggering until a set interval passes
- What you have built is a functional, scalable interactive scare system.

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SECTION 5

Scare Design Thinking

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TIMING AND ANTICIPATION

- The effect itself is only part of the experience — what happens before it matters equally
- Darkness, sound design, and guest pacing through the space prime the audience before any sensor activates
- In the Haunted Smelter: the physical environment built anticipation before the trigger fired
- The technology does not create the fear — it delivers it at the right moment

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RESET STATES AND COOLDOWN LOGIC

- An effect that retriggers immediately loses impact and breaks immersion
- In the Haunted Smelter, reset timing was a deliberate design decision, not an afterthought
- In TD: a Python timer in the DAT Execute prevents retriggering until the interval has passed
- Reset state: what does the system look like between guests — neutral video, ambient sound, base lighting

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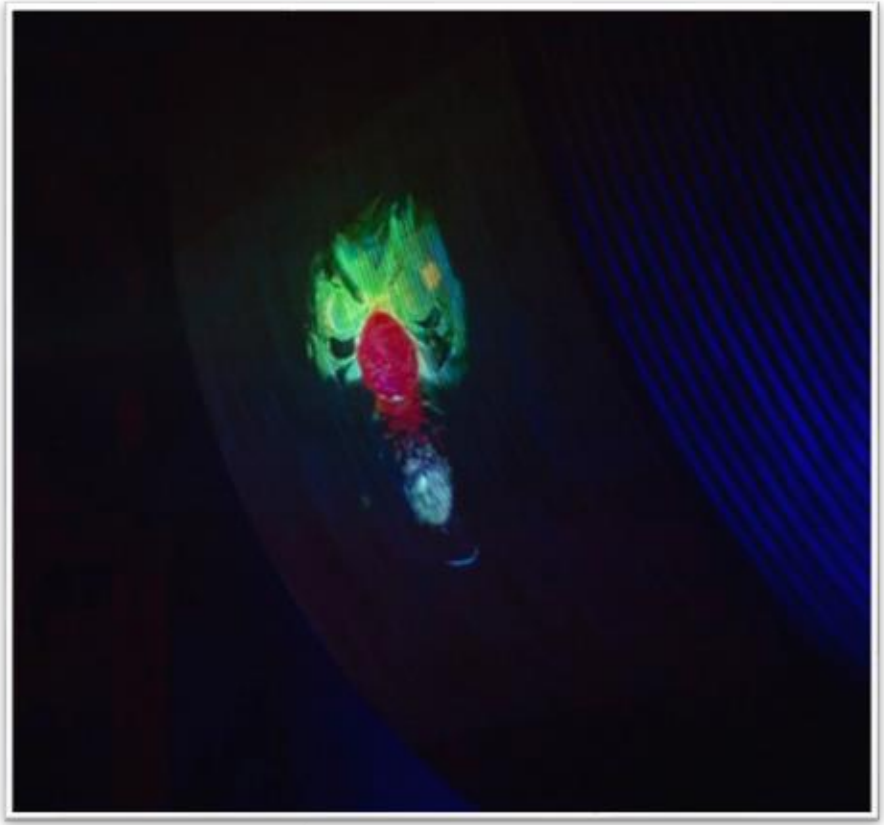
LAYERING MULTIPLE EFFECTS

- A single output — video only, or sound only — is less effective than synchronized layers
- In the Haunted Smelter: projection, sound, and lighting fired as one event
- Each layer reinforces the others — removing any one weakens the overall effect
- The shared output architecture in today's build keeps all layers synchronized from a single trigger



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SCALING FROM WORKSHOP TO FULL ATTRACTION

- Today's build is one trigger point. A full attraction chains many of these together
- **Scaling considerations:**
  - More inputs require more careful serial data management
  - More outputs require more organized network structure in TD
  - Core architecture stays the same: serial in → Python → CHOPs/DATs → TOPs/OSC
- ***What you built today is one room. A full attraction is many rooms using the same logic.***

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SECTION 6

Webcam Bonus Demo

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AN ALTERNATIVE INPUT PATH

The output logic is identical — only the input source changes.

Arduino Path	Webcam Path
• Physical sensors • Requires hardware • Standard in attractions • Used throughout today	• Camera-based detection • No additional hardware • Flexible placement • Explored now

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LIVE DEMO: MOTION DETECTION THRESHOLD

- Video Device In TOP → image processing TOPs isolate motion → CHOP extracts motion value
- Python script mirrors the logic from the build — threshold crossed, output fires
- Same output layer used — no changes downstream

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WHERE TO GO FROM HERE

Blob Tracking

Precise detection of individual bodies in the frame — useful for multi-person interactions

Optical Flow

Detects direction and speed of motion — enables more nuanced reactive systems

Derivative Documentation

docs.derivative.ca — comprehensive reference for all operators and Python API

TouchDesigner Forum

forum.derivative.ca — active community with examples and troubleshooting

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SECTION 7

Q & A

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QUESTIONS?

Resources available at the link below – includes the starter TD file, Arduino sketch, and wiring diagrams

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